MIL-C-11733D
11 April 1974
SUPERSEDING
MIL-C-11733C
27 September 1962

MILITARY SPECIFICATION

CYLINDER, COMPRESSED GAS: AMMONIA,

WITH VALVE (DOT 4AA480)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a new steel cylinder, with valve and cap, for anhydrous ammonia service.

1.2 Classification. The classification for the cylinder shall include the base specification letter and number, and the classification number from Table I as specified (see 6.2).

TABLE I.

CLASSIFICATION	CAPACITY POUNDS	D INCHES ±1/2	H INCHES ± 1	MAX. TAPE WEIGHT POUNDS	WATER CAPACITY CU. IN.	VALVE MIL-V-2/8
C-11733-1	100	15	38	105	5,140	V8-240-1
C-11733-2	150	15	53	162	7,700	V8-240-1

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-V-2 - Valves, Cylinder, Gas (for Compressed or Liquefied Gases), General Specification for.

MIL-V-2/8 - Valve, Cylinder, Gas: Anhydrous Ammonia (Convex-Head Cylinder) Outlet 241, Inlet 3/4 Inch.

FSC 8120

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MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-17376/1	- Caps and Flanges, Compressed Gas Cylinder: Caps.
MIL-C-17376/2	- Caps and Flanges, Compressed Gas Cylinder: Flanges, Low Pressure.
MIL-C-52752	- Cylinders, Compressed Gas, Packaging of.

STANDARDS

Military

MIL-STD-101	- Color Code for Pipelines and for Compressed Gas Cylinders.

DRAWING

BUREAU OF SHIPS

810-1385867	- Decalcomania for Navy Gas Cylinders.

(Copies of specifications, standards, and the drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION

Code of Federal Regulations, Title 49, DOT 4AA480.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. The ammonia cylinder shall conform to the Code of Federal Regulations DOT 4AA480. The cylinders shall be within dimensions specified in Table I, in accordance with Figure 1, and as specified herein.

3.2 Construction. The cylinders shall be true of form and shall be stable when in an upright position. Welded seams shall conform to DOT 4AA480. The footrings shall be attached by welding and shall have a minimum of [illegible] tion holes and two drain holes equally spaced around the base of the cylinder and shall be of the same outside diameter as that of the cylinder wall

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3.3 Cylinder processing.

3.3.1 Internal surfaces. The internal surfaces of drawn cylinders, prior to necking, shall be free of burrs, oxidation products, and foreign matter. When required, only iron-base abrasives shall be used to remove unstable surfaces or oxidation products from the cylinder wall. Metal discoloration, rust bloom, and fixed scale shall not be considered cause for rejection in steel cylinder fabrication.

3.3.2 Cylinder cleaning. After necking, drilling, threading, and deburring, the cylinders shall be cleared of loose particles by vacuum or by inverting the cylinder and the use of a compressed air sweep (see 6.3). The cylinder shall be flushed with hot fluids when necessary to remove grease, oil, and fat from processing. Chemicals that will react with anhydrous ammonia shall not be used.

3.3.3 Cylinder drying. Internal cylinder moisture from processing or hydrostatic test shall be removed by inverting the cylinder and drying with heated filtered air having a dew-point less than 30° F. In lieu of filtered air, drying by suction of ambient air through the cylinder is allowed. Closure shall be carried out immediately to limit atmospheric interchange.

3.4 Closure. Each cylinder shall be fitted with a steel flange in accordance with MIL-C-17376/2. Each flange shall be drilled and threaded 3/4 NGT and fitted with a valve in accordance with MIL-V-2/8 and as specified in Table I. Each cylinder shall be supplied with a valve protection cap in accordance with MIL-C-17376/1.

3.5 Tare weight. The tare weight of the cylinder shall be the weight of the cylinder, flange, and valve, but excluding the valve protection cap. The tare weight shall be expressed in pounds and 4 ounce increments to the nearest quarter pound. The figures expressing pounds shall be separated from the figures expressing ounces by a dash (-).

3.6 Test date. The cylinder shall be delivered within 1 year of the test date stamped on the cylinder.

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3.7 Cylinder identification. Unless otherwise specified herein, markings shall be in characters not less than 5/16 inches high.

3.7.1 Standard markings. Standard marking shall be plainly and permanently marked by stamping on the shoulder of each cylinder starting near the neck, as follows:

(a) "DOT4AA480"
(b) Serial number (see 6.4), as registered with the Bureau of Explosives in characters not less than 3/8 inch high for cylinders more than 6 inches in outside diameter.
(c) "USGOVT" Government symbol as registered with the Bureau of Explosives.
(d) Date of test diametrically opposite the above marking.
(e) The inspectors official mark.
(f) Manufacturer's or retester's mark near date of test, preferably right above it.
(g) Tare weight. The tare weight (see 3.5) shall be stamped in a position other than in sequence with the serial number or test date, and shall be preceded by the letters "TW".

3.7.2 Additional markings. In addition to the standard markings, the cylinder shall bear the following markings:

(a) The name of the gas stenciled on the cylinder in accordance with MIL-STD-101.
(b) When specified (see 6.2), decalcomania in accordance with Drawing 810-1385867 shall be affixed diametrically opposite one another on the sides of the cylinder, 90 degrees from the stenciled name of the gas.

3.8 Treatment and painting. Each cylinder and cap shall be treated and painted externally in accordance with MIL-T-704, Type A, color conforming to the cylinder color code requirements of MIL-STD-101.

3.9 Workmanship. Cylinders, valves, plugs, flanges, and caps shall be clean and free from grit, pits, and loose scale. Edges shall be rounded or chamfered. Cylinders shall be cleaned and free of dents, scratches, and any other surface defects detrimental to the intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Examination. Each cylinder shall be examined in accordance with 4.4.1 The presence of one or more defects shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. Cylinders shall be examined as specified herein for the following defects:

101. Cylinders not in accordance with DOT 4AA480, Code of Federal Regulations.
102. Dimensions not as specified.
103. Construction not as specified.
104. Cylinder processing not as specified.

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105. Closure not as specified.
106. Tare weight not as specified.
107. Test date more than 1 year prior to date of acceptance by the Government.
108. Marking not as specified.
109. Treatment and painting not as specified.
110. Workmanship not as specified.

4.5 Inspection of preparation for delivery. The packing and marking of the cylinders shall be examined in accordance with the Quality Assurance Provisions of MIL-C-52752.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. The cylinders shall be packed and marked in accordance with MIL-C-52752. The level of packing shall be Level A, B, or C as specified (see 6.2). Cylinders shall be packed as specified for Type VI.

6. NOTES

6.1 Intended use. These cylinders are intended for use in shipment and storage of anhydrous ammonia.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Cylinder classification number required (see 1.2).
(c) When decalomanias are required (see 3.7.2).
(d) Level of packing required (see 5.1).

6.3 Loose particulate matter in stored cylinders. Cylinders that have been cleaned, dried, valved or plugged and subjected to storage and handling within delivery time limits will be acceptable for ammonia service if accumulated loose ferro-related particulate matter resulting from handling does not exceed 1 gram per 1500 cubic inches of cylinder capacity.

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6.4 Serial numbers. Cylinder serial numbers shall be prefixed by a two or three letter symbol designating the procuring agency and shall have a two letter suffix designating the supplier furnished by the contracting officer. The serial numbers shall be assigned by the supplier. These numbers may be consecutive with the supplier's regular production numbers or of a series established specifically for customer cylinders. However, all cylinders on a given contract will be numbered consecutively and controls will be exercised to preclude duplication on future deliveries to the Government.

Custodians:

Army - ME
Navy - SH
Air Force - 68

User activities:

Navy - MC, YD

Preparing activity:

Army - ME

Project No. 8120-0337

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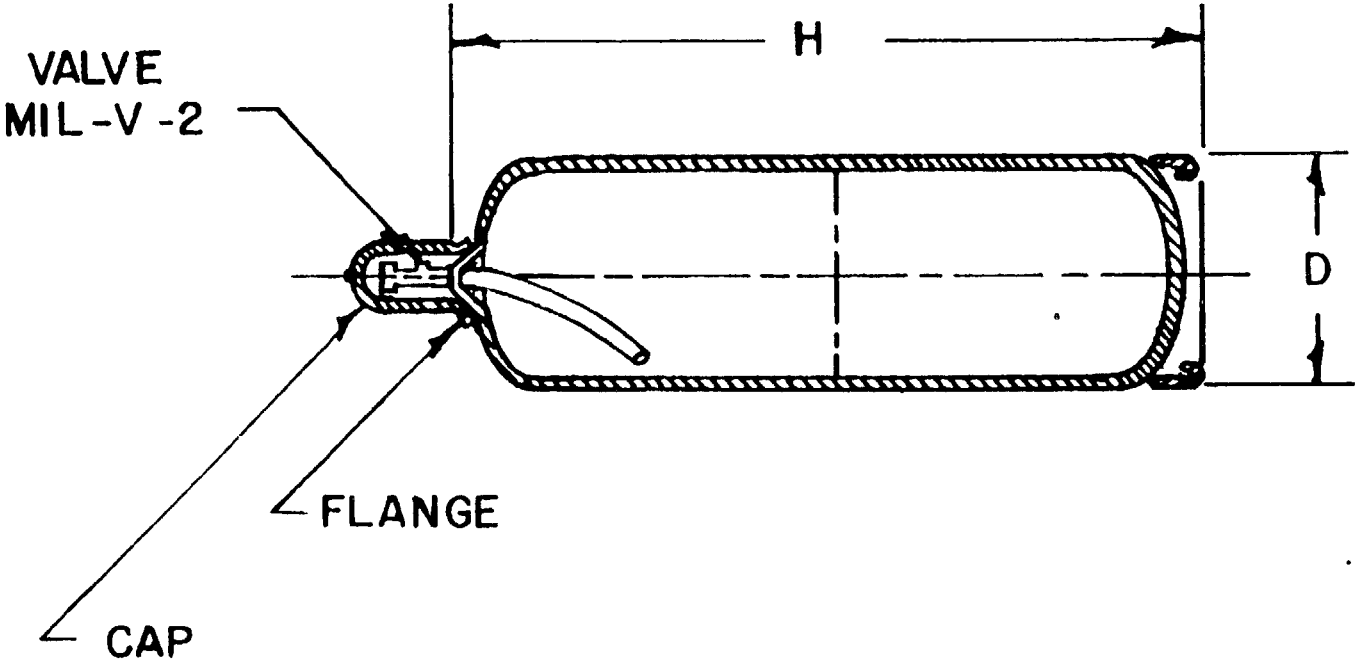


FIGURE 1. AMMONIA CYLINDER 4AA480
X-2461

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-C-11733D, Cylinder, Compressed Gas· Ammonia, with Valve (DOT 4AA480)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes" in what way?)

4 REMARKS (Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity Optional)

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES PREVIOUS EDITION OF 1 OCT 64 WHICH MAY BE USED

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